

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Hotel Billing & Restaurant Management System

Version: 1.0

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1. Introduction

1.1 Purpose

The purpose of this project is to develop a cloud-based Hotel Billing and Restaurant Management System that digitizes restaurant operations including order management, kitchen processing, billing, reporting, and staff management.

The system will provide real-time communication between waiters, chefs, and managers through mobile and web applications.

1.2 Scope

The system will:

- Manage restaurant orders digitally.
- Generate bills automatically.
- Provide real-time order updates.
- Allow managers to monitor operations remotely.
- Store data securely in the cloud.
- Support multiple user roles.
- Generate reports and analytics.

The system is intended for restaurants, hotels, cafes, and food service businesses.

1.3 Project Goals

- Eliminate paper-based ordering.
- Improve communication between waiters and kitchen staff.
- Reduce billing errors.
- Increase operational efficiency.
- Provide centralized cloud-based management.
- Build a scalable system for future expansion.

2. System Overview

The system consists of three applications:

2.1 Waiter Mobile Application

Used by waiters to:

- Login
- View menu
- Create orders
- Modify orders
- Send orders to kitchen
- Track order status

2.2 Chef Application

Used by kitchen staff to:

- Receive incoming orders
- View order details
- Update preparation status
- Mark orders as ready

2.3 Manager Dashboard

Used by managers to:

- Monitor restaurant operations
- Manage menu items
- Generate bills
- View reports
- Manage employees
- Access analytics

3. User Roles

3.1 Administrator

Responsibilities:

- Manage system users
- Configure system settings

- Assign user roles
 - Access all system modules
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3.2 Manager

Responsibilities:

- View dashboard
 - Manage menu
 - Generate bills
 - Access reports
 - Manage restaurant operations
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3.3 Waiter

Responsibilities:

- Create customer orders
 - Update customer orders
 - View assigned tables
 - Track order status
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3.4 Chef

Responsibilities:

- View pending orders
 - Update cooking status
 - Complete orders
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4. Functional Requirements

4.1 Authentication Module

Features

- User Login
- User Logout
- Password Encryption
- JWT Authentication
- Role-Based Access Control

Acceptance Criteria

- Only registered users can access the system.
 - Users can only access features permitted by their role.
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4.2 Menu Management Module

Features

- Add Menu Item
- Edit Menu Item
- Delete Menu Item
- View Menu Items
- Categorize Menu Items

Acceptance Criteria

- Managers can create and update menu items.
 - Menu changes should be visible to all applications.
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4.3 Table Management Module

Features

- Create Tables
- View Table Status
- Update Table Availability

Table Status

- Available
 - Occupied
 - Reserved
 - Closed
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4.4 Order Management Module

Features

- Create Order
- Modify Order
- Cancel Order
- View Order History
- Track Order Status

Order Status

- Pending
- Accepted
- Preparing
- Ready
- Delivered
- Completed
- Cancelled

Acceptance Criteria

- Orders must be stored immediately.
 - Orders should be visible to kitchen staff instantly.
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4.5 Kitchen Management Module

Features

- View Incoming Orders
- Update Order Status
- Mark Order Ready
- View Completed Orders

Acceptance Criteria

- Kitchen staff must receive orders in real time.
 - Order updates must be visible to waiters instantly.
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4.6 Billing Module

Features

- Generate Bill
- Calculate Taxes
- Apply Discounts
- Print Receipt
- Download PDF Invoice

Payment Modes

- Cash
- UPI
- Credit Card
- Debit Card

Acceptance Criteria

- Bills should be generated accurately.

- Taxes and discounts should be calculated automatically.
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4.7 Reporting Module

Features

- Daily Sales Report
- Weekly Sales Report
- Monthly Sales Report
- Revenue Analysis
- Best Selling Items

Acceptance Criteria

- Reports should be exportable.
 - Reports should display accurate sales information.
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4.8 Employee Management Module

Features

- Add Employee
- Update Employee
- Remove Employee
- Assign Roles

Employee Roles

- Admin
 - Manager
 - Waiter
 - Chef
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5. Non-Functional Requirements

5.1 Performance

- System should respond within 2 seconds.
 - System should support at least 50 concurrent users.
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5.2 Availability

- System should be available 24/7.
- Cloud infrastructure should ensure high availability.

5.3 Security

- Passwords must be encrypted.
- Authentication must use JWT.
- Role-based authorization must be implemented.
- Sensitive data must be protected.

5.4 Scalability

- System should support future branch expansion.
- System should support additional modules.

5.5 Reliability

- Data should not be lost during failures.
- Automatic backups should be available.

6. System Architecture Requirements

Frontend

Waiter App

Technology:

- Android Studio
- Java
- Material Design

Chef App

Technology:

- Android Studio
- Java

Manager Dashboard

Technology:

- React
- TypeScript
- Tailwind CSS

Backend

Technology:

- Java
 - Spring Boot
 - Spring Security
 - Spring Data JPA
 - Spring WebSocket
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Database

Technology:

- PostgreSQL
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Cloud Services

AWS Services:

- EC2
 - RDS PostgreSQL
 - S3
 - CloudWatch
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DevOps

Tools:

- Git
 - GitHub
 - Azure DevOps Pipelines
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7. Future Enhancements

- QR Code Ordering
- Inventory Management
- Customer Loyalty Program
- Online Payments
- SMS Notifications
- Email Notifications
- Multi-Branch Support
- AI-Based Sales Prediction

- AI-Based Demand Forecasting
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8. Success Criteria

The project will be considered successful if:

- Waiters can create and manage orders.
 - Chefs receive and process orders.
 - Managers can generate bills and reports.
 - Data is accessible from anywhere through cloud infrastructure.
 - Real-time updates function correctly.
 - System remains secure and reliable.
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9. Conclusion

The Hotel Billing and Restaurant Management System aims to provide a modern cloud-based solution for managing restaurant operations. The system will improve efficiency, reduce manual effort, and provide a scalable platform capable of supporting future business growth.